

Release: KK PI 26.4.22

Release ID: #318103

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Branch: release/318103_KK_PI_26.4.22

1. Summary

- **Claims engine:** Error 76 infant-disallowance logic added for HX030 (J.017 policy), Child Reconciliation report clears DO/PU instead of In/Out times (#319145 regression fix), OER R/I/R,I character correction for infants, HX211 snack-capacity formula fix, Full Month Attendance ISNULL wrap for missing child names.
- **Meal Counts with FRP:** New policy L.11 (INCLUDE_ALL_RECORDED_MEALS_FRP_FLAG) lets a client opt in to include all recorded meals regardless of disallowance (#318686).
- **Sponsor 102 Print Check:** Cross-repo print pipeline overhaul — new Sponsor102Data (Active Child List, tier, training hours, meal times) flows through HX API → KK → PDF; new ACH re-download without void/reissue; new CHECK_REGISTER covering index (#316306).
- **J.017 policy (HX030 own-infant edit-check):** New SPONSOR_POLICY.include_infants_in_own_alone_edit_check_flag column + HX API entity + VB6 property + HX Desktop v3.4.0 build; seeded Y for sponsor 030 only (#320244).
- **State-specific tiering:** New STATE_POLICY.use_school_tier_exact_date_flag set Y for Iowa (state 20) — allows custom school tier start/end dates (#319205).
- **Enrollment / eForm:** New admin endpoints common/detectOrphanedEforms and common/cleanupOrphanedEforms (security-key auth) to purge orphaned CX eForm assignments; CX Activate Children null-enrollment fix (#308617); home-sponsor-observer can add parents (#318629).
- **Reports & Texas IEF:** TX Child-Care IEF instruction pages removed (#320684); TX Adult-center IEF now correctly routed via program_type_code (#321830); select_rpt_401_child_enrollment_sp updated.
- **Spanish localization:** Added translations for Home Provider flows (#298391) — partial coverage, some keys still English (see Risk).
- **Provider policy defaults:** highest_serving_allowed_code defaults to 215 (Serving 1) when saved as 0/NULL (#320881).
- **Data patch:** Restore foster-child historical IE dates wiped by PI #318945 (#321413, HX102 request).
- **Parachute (bundled):** Standardized search/clear-X across accounting pages, kiosk PIN/phone numpad responsive, number-input caret fix, State Agency split-amount zero-out fix.

2. Infrastructure Impact

Overall Risk Level: High

2.1 Database Changes

- MMADMIN/Updates/319205_Add_use_school_tier_exact_date_flag.sql — adds use_school_tier_exact_date_flag column to STATE_POLICY , sets 'Y' for state_code=20 (Iowa).
- MMADMIN_XXX/Updates/320244_ADD_NEW_FIELDS_TO_TABLE_SPONSOR_POLICY.sql — adds include_infants_in_own_alone_edit_check_flag to SPONSOR_POLICY (NOT NULL DEFAULT 'N').
- MMADMIN_XXX/Updates/320244_SET_DEFAULT_FIELDS_TO_TABLE_SPONSOR_POLICY.sql — backfills existing rows to 'N'.
- MMADMIN_XXX/Updates/320244_SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql — sets 'Y' for sponsor scan_code='030'. **[VERIFY — currently NOT included in regenerated MRP.sql manifest; requires manual DBA run, see section 2.4]**
- MMADMIN_XXX/Updates/321413_Restore_Foster_Child_Income_Eligibility.sql — one-off data patch to restore foster children's historical IE dates from CHILD_TIER_BACKUP_318945 .
- MMADMIN_XXX/Programmability/Stored_Procedures/save_sponsor_policy_sp.sql — new trailing parameter @include_infants_in_own_alone_edit_check_flag flag = 'N' .
- MMADMIN_XXX/Programmability/Stored_Procedures/save_provider_policy_sp.sql and save_provider_policy_sp_port_kk.sql — 215 default applied when input is 0 or NULL.
- MMADMIN_XXX/Programmability/Functions/claim_error_meal_attendance_count_fn.sql — full rewrite; Error 76 branch adds CLAIM_ERROR→PROVIDER→SPONSOR_POLICY join per invocation to read J.017.

- MMADMIN_XXX/Programmability/Functions/claim_error_meal_attendance_source_form_fn.sql — adds MEAL_ATTENDANCE scan for source_form_code 151/152 to return 'I', 'R', or 'R,I'.
- MMADMIN_XXX/Programmability/Stored Procedures/select_361126_full_month_attendance_worksheet_sub1_sp.sql — wraps child name columns with ISNULL(..., '') .
- MMADMIN_XXX/Programmability/Stored Procedures/select_361127_Child_Reconciliation_sp.sql and CXADMIN equivalent — adds second UPDATE on #PGRID to clear DO/PU on RECTYPE='A' rows (fix for #319145).
- MMADMIN_XXX/Views/STATE_POLICY.sql and STATE_POLICY_TD.sql — rebuilt to expose new state flag column.
- CXADMIN/Updates/316306_create_check_register_client_center_covering_index.sql — new non-clustered covering index IX_CHECK_REGISTER_client_id_center_id ON CHECK_REGISTER with 7 INCLUDE columns. **Rollback script exists.**
- CXADMIN/Updates/318686_add_new_policy_L11_include_all_recorded_meals_frp.sql — inserts policy L.11 into POLICY . Default L.11 CLIENT_POLICY value is effectively 'N' via ISNULL fallback (no CLIENT_POLICY row inserted by this release).
- CXADMIN/Programmability/Stored Procedures/select_payment_history_list_sp.sql — emits 3 new columns: mod_date_time , mod_login_id , mod_username . Consumed by CX LogicService + KK AutoMapper.
- CXADMIN/Programmability/Stored Procedures/select_rpt_meal_count_attendance_sp.sql — flips @respectDisallowance default 1→0; policy-L11-driven. Fallback ISNULL preserves legacy behavior when CLIENT_POLICY row absent.
- CXADMIN/Programmability/Stored Procedures/select_rpt_401_child_enrollment_sp.sql — blank IEF branch now uses ISNULL(CLC.program_type_code, 0) instead of hardcoded 0 (supports TX adult/child IEF split).

2.2 Auth / Session Changes

- None
- Expected re-authentication impact: None — no auth changes.

2.3 API Changes

- NEW: POST /common/detectOrphanedEForms (security-key auth) — enumerates CX clients with orphaned eForm assignments. Payload: { SecurityKey, ClientIds?[] } .
- NEW: POST /common/cleanupOrphanedEForms (security-key auth) — hard-deletes orphaned CX eForm assignments. Payload: { SecurityKey, ClientIds?[] } . **When ClientIds is null/empty, the endpoint sweeps every CX client with any active assignment.** [VERIFY — see section 2.4.]
- NEW: POST /portcx/payment/redownloadACHFile — regenerate ACH file for a set of check_ids without voiding. Payload: { CheckIds: int[] } (permission-gated to CX.IssuePayments).
- NEW: GET /site/isAtRiskOrArasForSponsor?centerId=... — returns live at-risk/ARAS flag for a center. **[VERIFY — endpoint added but call site not visible in diff; confirm it is consumed.]**
- CHANGED: POST /portcx/fileservice/updateIEFChildren — return shape changed from bool to { success, skippedChildNames[] } .
- CHANGED: GET /cxservice/payments/getCheckRegisterList — response now includes modDateTime , modLoginId , modUsername on each row.
- CHANGED: POST /payments/printIssuePayment (HX API) — response now includes Sponsor102Data { ActiveChildren[], ChildTierCounts[], ProviderTiers[], ServingTimes[], TrainingHours } for Sponsor 102 providers.
- CHANGED: HX API SponsorPolicy entity — new property include_infants_in_own_alone_edit_check_flag . HX API StatePolicy DTO (via SelectStatePolicyDTO) — new use_school_tier_exact_date_flag .

2.4 Performance & Capacity

- **CHECK_REGISTER covering index (#316306)** — new non-clustered index with 7 INCLUDE columns on a multi-million-row table. CREATE NONCLUSTERED does not specify ONLINE = ON → Sch-M lock for duration of build, blocks all reads/writes to CHECK_REGISTER.
Action required — run DDL in maintenance window; add WITH (ONLINE = ON, MAXDOP = 4) if Enterprise Edition.
Rollback: Low — standalone ROLLBACK script exists at CXADMIN/ROLLBACK/316306_create_check_register_client_center_covering_index.sql .
Watch: CHECK_REGISTER write latency during and immediately after index creation; sys.dm_db_index_usage_stats 7 days after release to verify the index is used.
Evidence: Pattern #5 scope-creep risk — non-SARGable query Year(cr.check_date)=Year(@claim_month) in select_payment_history_list_sp still scans client-wide even with the covering index.
- **Orphan detect/cleanup endpoints (#319630)** — common/cleanupOrphanedEForms with auto-detect (null clientIds) performs serial per-client fan-out: opens fresh eFormsContext + CxContext per iteration and pulls entire active CHILD roster per client. No CommandTimeout . No row cap.
Action required — (1) require non-empty clientIds in prod (reject auto-sweep), (2) add explicit Database.CommandTimeout = 30-60s , (3) add [RequiredRole] on top of security-key check, (4) firewall to DevOps app pool / IP allow-list, (5) audit-log every invocation (caller, ClientIds, row counts).
Rollback: Low — disable endpoint via IIS route rule or back-out code.
Watch: HTTP request duration on /common/cleanupOrphanedEForms ; SQL connection count against CXADMIN and EFormContext during first sweep;

KK API worker-thread count.

Evidence: Matches known incident patterns 3/4 (connection accumulation, thread-pool starvation, elevated CommandTimeout). **Also: orphan predicate treats Withdrawn/Inactive children as orphans → hard-deletes their historical eForms. Recommend NOT running in auto-sweep mode on prod until predicate is tightened.**

- **Sponsor 102 PrintIssuePayment fan-out (#320213)** — HX API controller uses unbounded `Task.WhenAll` calling `BuildSponsor102Data` per check; each call fires 5 stored procedures in parallel. A 500-provider batch print = 2,500 concurrent DB round-trips.

Monitor — spike in MMADMIN connection count during first post-release 102 print.

Rollback: Medium — requires HX API redeploy to revert.

Watch: `sp_who2 / sys.dm_exec_connections` during first 102 batch print; HX API p99 latency on `/payments/printIssuePayment`.

Evidence: No SemaphoreSlim, no caching across providers sharing tier/fiscal data.

- **claim_error_meal_attendance_count_fn Error 76 branch (#320244/#321350)** — adds `CLAIM_ERROR→PROVIDER→SPONSOR_POLICY` join per invocation. Scalar function called per `CLAIM_ERROR` row in OER/summary reports.

Monitor — execution time of `select_claim_error_summary_sp` and `select_claim_errors_for_rpt_sp` post-deploy for sponsors with high Error 76 volume.

Rollback: Low — revert function via `CREATE OR ALTER`.

Watch: Report query duration on OER for HX030 and HX102.

Evidence: Pattern #1-style transformation from pre-resolved flag to per-row lookup.

- **ACH re-download CommandTimeout = 300s (#316306)** — `GetCenterACHFileFromCheckRegister` sets 5-minute command timeout for an indexed IN-lookup.

Monitor — occurrences of 5-minute SQL waits on `check_register`.

Rollback: Low — code redeploy.

Watch: SQL query execution time on ACH redownload calls.

Evidence: Pattern #4 anti-pattern (oversized timeouts mask real issues).

- **MRP_Rollback scope gap** — `CXADMIN/MRP_Rollback.build.config` only covers `select_payment_history_list_sp`. Rollbacks for 318686 policy `INSERT`, `select_rpt_meal_count_attendance_sp`, `select_361127_Child_Reconciliation_sp`, `select_rpt_401_child_enrollment_sp` are NOT in the bundled rollback.

Action required — DBA must hand-craft rollback for those SPs + `DELETE` policy row if rollback is triggered.

Rollback: Medium — partial automated, partial manual.

Watch: N/A (procedural).

Evidence: Manifest diff inspection.

- **MRP missing sponsor-030 enablement (#320244)** — `320244_SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql` exists in repo but is NOT in the regenerated MRP manifest. Feature will not activate for HX030 unless DBA runs manually.

Action required — either regenerate MRP with the script included (sequenced after `SET_DEFAULT`), OR document manual DBA run post-deploy with product-owner sign-off.

Rollback: Low —

```
UPDATE SPONSOR_POLICY SET include_infants_in_own_alone_edit_check_flag='N' WHERE sponsor_id = (SELECT sponsor_id FROM SPONSOR WHERE scan_code='
.
```

Watch: HX030's Error 76 behavior post-deploy — if no change observed, the script was not run.

Evidence: MRP manifest inspection.

2.5 Cross-Repo Dependencies

- Repos involved: **MinuteMenu.Database**, **POC_HxCloudConnectApi** (HX API), **HX** (Desktop/VB6 — build 3.4.0), **Centers-CX**, **Shared-Core**, **KK**, **Parachute**
- Deploy order: **MinuteMenu.Database** → **Shared-Core** → **POC_HxCloudConnectApi** → **HX Desktop (3.4.0)** → **Centers-CX** → **KK** → **Parachute**
 - Rationale: DB schema change must precede any EF-mapped reader (HX API `SponsorPolicy` entity); HX Desktop v3.4.0 reads the new column unconditionally and will fail sponsor-policy load on a pre-upgrade DB; KK consumes the updated CX `MinuteMenu.Centers.ServerLib.dll` plus SP signature changes.
- Rollback order: **Parachute** → **KK** → **Centers-CX** → **HX Desktop** → **POC_HxCloudConnectApi** → **Shared-Core** → **MinuteMenu.Database**

3. Web Configuration Changes

Service	Key	Value	Notes
HX API	(none per diff)	—	settings.xml changed but no new key surfaced in review — [VERIFY]

Service	Key	Value	Notes
KK BE	(none)	—	No new appsettings key added in this release

[VERIFY] — Confirm whether `HxCloudConnect.Api/settings.xml` changes require corresponding prod-environment config update.

4. Rollback Plan

1. **Halt further traffic to printIssuePayment endpoint** (Sponsor 102 print) if rollback is decided mid-day to avoid incomplete PDF batches.
2. Deploy previous Parachute build.
3. Deploy previous KK BE + FE build.
4. Deploy previous Centers-CX build (restores the pre-change `MinuteMenu.Centers.ServerLib.dll`).
5. Reinstall prior HX Desktop installer (pre-3.4.0) on affected sponsors — **HX Desktop rollback is per-machine; communicate with affected sponsor operators.**
6. Deploy previous HX API build.
7. Deploy previous Shared-Core build (reporting service file replacement).
8. Run DB rollback scripts:
 - `CXADMIN/ROLLBACK/316306_create_check_register_client_center_covering_index.sql` (drops covering index)
 - `CXADMIN/RELEASES/MRP_Rollback.sql` (restores `select_payment_history_list_sp`)
 - **Manual rollback required** for: `select_rpt_meal_count_attendance_sp`, `select_361127_Child_Reconciliation_sp`, `select_rpt_401_child_enrollment_sp`, `claim_error_meal_attendance_count_fn`, `claim_error_meal_attendance_source_form_fn`, `select_361126_full_month_attendance_worksheet_sub1_sp`, `save_sponsor_policy_sp`, `save_provider_policy_sp`, `save_provider_policy_sp_port_kk` — DBA to check out prior commits from master and re-deploy each SP.
 - `DELETE FROM POLICY WHERE policy_name='INCLUDE_ALL_RECORDED_MEALS_FRP_FLAG'` and any related `CLIENT_POLICY` rows.
 - `UPDATE SPONSOR_POLICY SET include_infants_in_own_alone_edit_check_flag='N'` (reset J.017 to legacy) — or, if full column removal is required, `ALTER TABLE SPONSOR_POLICY DROP COLUMN include_infants_in_own_alone_edit_check_flag`.
 - `ALTER TABLE STATE_POLICY DROP COLUMN use_school_tier_exact_date_flag` — remove state tier flag.
 - `321413_Restore_Foster_Child_Income_Eligibility.sql` cannot be rolled back directly — foster IE dates written back into `CHILD` are correct by intent. If reversal is needed, restore `CHILD` rows from `CHILD_TIER_BACKUP_318945` for the affected rows.
9. Post-rollback: run HX030 Error 76 regression test and an OER smoke for one large sponsor to confirm claim-processing pipeline is back to pre-release behavior.

5. Risk Assessment

- **Worst case:** (1) HX API outage on production if deploy order is reversed — the EF-mapped `SponsorPolicy` entity reads a column that doesn't exist yet; every route using `SponsorPolicies` breaks. (2) Mass eForm data loss if the cleanup endpoint is invoked in auto-sweep mode against the current orphan predicate (withdrawn-children = orphan). (3) Silent overwrite of `PROVIDER_POLICY.highest_serving_allowed_code` to 215 on every HX Desktop save → incorrect meal allowance on next claim processing for any provider configured to Serving 2/3.
- **How we detect it:** (a) APM error rate on HX API for "Invalid column name 'include_infants_in_own_alone_edit_check_flag'" within 5 min of HX API deploy; (b) APM on `/cxservice/payments/getCheckRegisterList` for `IndexOutOfRangeException` if CX/KK deploys before DB; (c) sudden drop in `PROVIDER_POLICY.highest_serving_allowed_code != 215` count over a 24h window on HX030/HX102; (d) spike in `EnrollmentAssignmentCx` hard-delete rate (DB audit or `sp_BlitzCache "Deletes per second"` counter).
- **How we recover:** Follow Rollback Plan section 4. For (c), restore `PROVIDER_POLICY` from nightly backup. For (d), restore `EnrollmentAssignmentCx`, `EnrollmentForm`, `IncomeEligibilityForm`, and related storage from nightly backup; notify affected clients; re-enter lost eForms if recovery is incomplete.

6. Deployment Checklist

#	Step	Owner	Status
1	Dev lead prepared release note	Tuan Nguyen	Done

#	Step	Owner	Status
2	DevOps reviewed infrastructure impact (esp. deploy order + orphan cleanup endpoint risk)	[VERIFY — name]	Pending
3	Architect reviewed changes (J.017 cross-repo contract + Sponsor 102 fan-out)	[VERIFY — name]	Pending
4	SQL scripts verified on staging (MRP + 320244 set + 321413 + 316306 index)	[VERIFY — name]	Pending
5	Web config changes documented and ready	[VERIFY — name]	Pending
6	Rollback plan confirmed (including manual SP rollback steps)	[VERIFY — name]	Pending
7	Confirm 320244_SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql added to MRP OR scheduled for manual run	[VERIFY — name]	Pending
8	Verify CHILD_TIER_BACKUP_318945 exists on each HX sponsor DB before running 321413	[VERIFY — name]	Pending
9	316306 index DDL scheduled for maintenance window OR confirmed Enterprise Edition for ONLINE=ON	[VERIFY — name]	Pending
10	HX Desktop 3.4.0 rollout plan coordinated with sponsor support (per-machine installer)	[VERIFY — name]	Pending
11	Sponsor 102 print-batch SQL connection monitor staged	[VERIFY — name]	Pending
12	Cleanup-orphaned-eForm endpoint firewalled OR code-gated before production access	[VERIFY — name]	Pending
13	Production deployment approved	[VERIFY — name]	Pending

7. Tickets

Ticket	Title	Type	State	Infra Impact
#296656	[Sponsor] Run button disabled switching SFSP & ARAS	Story	Withdrawn	None
#297720	[MPR ARAS/SFSP] Rqd Serving Size not displayed	Story	Withdrawn	None
#298391	Add Spanish translation to KidKare for Home Providers	Story	Ready for Release	Low
#299748	KCE: At Risk Portion Of Child Roster Missing	Story	UAT	Low
#300571	CMK: Diet Statement on File Diet Exp Date disabled	Story	Withdrawn	None
#300738	IL State Provider File incorrect Begin Date	Story	Withdrawn	None
#300918	Copy options not remembered Infant/Non-Infant tabs	Story	Ready for Release	Low
#308617	[CX] Error when Set IEF with NULL Current Enrollment	Story	UAT	Low
#316306	KK Centers Re-Generate Payments Without Void/Reissue	Story	Ready for Release	High
#318117	Geo-Location Missing HX300 Reviews	Story	UAT	Low
#318178	Child Attendance Reconciliation overlapping DO/PU	Story	Ready for Release	Medium
#318581	Ghost eForm Enrollment Record (Not Started)	Story	Dev	[VERIFY — no code change]
#318584	Receipt Verification does not list receipts	Story	Code review	Low
#318629	Home sponsors cannot add parents when observing	Story	Ready for Release	Low
#318686	Meal Counts & Attendance With FRP missing meal data	Story	UAT	Medium
#318784	Child Roster licenseld override uses live At-Risk	Bug	UAT	Low
#319145	#318178 Fix clears In/Out instead of DO/PU on RECTYPE='A'	Bug	UAT	Medium
#319149	Center Search Not Functioning After Clear	Story	UAT	Low
#319156	Blank Attendance + Served Meals shows withdrawn children	Story	UAT	Low
#319205	HX100 School Start/End date update	Story	Ready for Release	Medium

Ticket	Title	Type	State	Infra Impact
#319407	Some children names missing on Full Month Attendance	Story	Ready for Release	Low
#319413	HX 102 Void payment affected all providers	Story	Analysis	[VERIFY]
#319630	CX client 8141: Child Not Found approving eForms	Story	Code review	High
#320213	[Sponsor 102 Print Check] Data not populated	Story	Test	High
#320224	Eliminate depreciatecx dependency — Move eForm/FRP to KK	Story	Dev	[VERIFY]
#320244	HX030 Error 76 disallow own infants	Story	Ready for Release	Critical
#320295	Need assistance tracking meals for Louisiana	Story	Withdrawn	None
#320305	#320224 - Specifications	Task	Open	None
#320443	A provider cannot enter meal	Story	Released	Low
#320684	Texas - Remove IEF instructions pages (Child Care)	Story	Ready for Release	Low
#320881	PCI Highest Serving Allowed empty	Story	UAT	Medium
#320883	Error 113 firing after PI318945 cleared child tier	Story	PDT Input	[VERIFY]
#320884	Actual Quantity Served of all foods in March removed	Story	Code review	Low
#320891	Cash In Lieu not calculated for KKFP client 3923	Story	Completed by Support	None
#321350	OER shows R character for Infant (J.017=Y)	Bug	UAT	Medium
#321398	Missing data in the OER report	Bug	UAT	[VERIFY — empty ticket description]
#321408	HX211 Provider Capacity snack meal counts	Story	Ready for Release	Low
#321413	Restore Historical IE Dates for Foster Children	Story	Test	Medium — data patch
#321481	Determine how March 2026 claim was deleted	Story	Support	None
#321485	Completed eForms not appearing under Reporting	Story	Withdrawn	None
#321486	Kidkare Slowness — pages not loading	Story	Released	None
#321487	Issue Payments Partial Posting Recurrence 4/3	Story	Released	None
#321662	Claim: Unable to disallow Pending children after Withdrawal	Story	Ready For Test	[VERIFY — no code change]
#321830	Texas - IEF for Adult center showing as IEF for child	Story	Ready for Release	Medium
#322138	Data different between KK and HX app (Print Check)	Bug	UAT	Medium

[VERIFY] items for dev lead:

- **#318581, #321662** — no code change found in any release diff. Confirm with engineering whether in scope.
- **#319413, #320224, #320305, #320883** — in Analysis/Dev/PDT Input/Open states; confirm inclusion or move out.
- **#321398** — empty ticket description; no code references. Confirm scope.